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Paleosecular variation from Easter Island revisited: modern demagnetization of a 1970s data set

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Abstract

Sixty-two Brunhes age lava flows from Easter Island in the Eastern Pacific Ocean, collected and processed for paleomagnetic and secular variation studies in the 1970s, have been reprocessed to meet current standards of demagnetization. Alternating field demagnetization in 7–10 steps was performed on all samples previously undemagnetized and characteristic directions determined from principal component analysis. The final data set gives a mean direction of $I = -46.1^\circ$ and $D = 357.4^\circ$ with $\alpha_{95} = 3.9^\circ$ Eliminating sites with 95% confidence circles greater than 10.0° gave an angular dispersion of virtual geomagnetic poles (VGPs) of 14.1° with respect to the rotation axis. This value is slightly larger than that determined prior to detailed demagnetization, and agrees with values predicted for this latitude by Model G. There is no compelling evidence from this study supporting the existence of a secular variation low in the southeastern Pacific region.
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1. Introduction

There has been controversy over the presence or absence of a secular variation low in the earth's magnetic field in the Pacific basin for some time, beginning with the work in Hawaii by Doell and Cox (1971, 1972), in which they suggest the presence of a non-dipole low in the central Pacific. These authors used lower than expected values of angular dispersion of virtual geomagnetic poles (VGPs) from Hawaiian lava flows to support this assumption. Comparisons made to angular dispersion from other locations around the world averaging similar Brunhes age populations reveal higher values of dispersion elsewhere and closer agreement with suggested global models for the variation of dispersion over latitude. Other investigations for paleosecular variation in the Pacific, including Easter Island (Isaacson and Heinrichs, 1976), the Galapagos Islands (Cox, 1971), New Zealand (Cox, 1969a) and the Aleutian Islands (Bingham and Stone, 1972) all show paleosecular variation values close to the models for their given latitudes.

Recent studies in the Pacific region have added to the central Pacific non-dipole low controversy, with individual studies supporting the presence of the low (McWilliams et al., 1982; Shibuya et al., 1995; Miki et al., 1998). On the other hand, McElhinny et al. (1996) make a case for the observed low dispersion being strongly influenced by data selection, and using three identified criteria find that Hawaiian data set yield expected dispersion relative to global averages.

Many early paleomagnetic studies used for secular variation studies have been criticized by McElhinny

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and McFadden (1997) on the basis of demagnetization procedures which do not measure up to present standards. They suggest that a number of studies, mostly done in the 1970s, need to be redone to provide reliable and defendable paleosecular variation data. These data are needed not only for investigating regional variations in the time averaged field, but also for constraints on global field models. In this paper, a large collection of lava flow data from Easter Island in the southeast Pacific sampled and measured in the early 1970s are reprocessed to investigate paleosecular variation during the Brunhes epoch.

2. Geologic setting

Easter Island (27.1°S, 109.2°W) is situated on the Nasca Plate, lying along the Easter/Sala y Gomez ridge, east of the East Pacific rise. It appears to be the younger end of the Easter Seamount Chain, extending east–west across the Nasca Plate. The island is separated from the East Pacific rise by the Easter microplate, a tectonic feature existing in the area for the past 5 million years (Naar and Hey, 1991; Cogne et al., 1995). Although, the origin of the seamount chain and the location of a supposed hot spot has been debated, recent work indicates that the plume is located near Sala y Gomez island, east of Easter Island by some 400 km (O'Connor et al., 1995). The sea floor beneath Easter Island records magnetic anomaly 3 (Naar and Hey, 1991), with a mean age of 3 million years.

The geology of Easter Island has been studied by Baker (1966), Baker et al. (1974) and more recently by Haase et al. (1997). The island has been divided into three volcanic subdivisions (Fig. 1) following three prominent volcanic centers: Poike, Rau Kau and

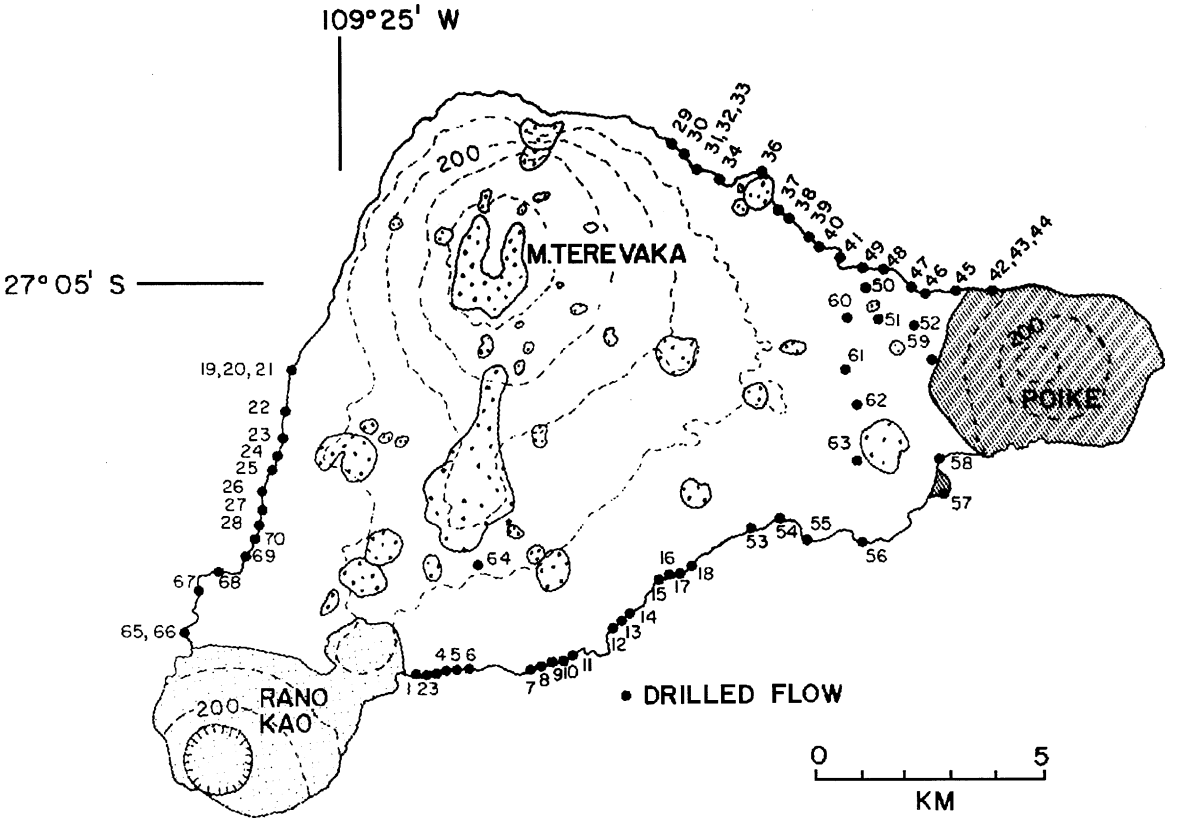


Fig. 1. Geology and site locations, Easter Island. Geology is from Baker (1966) and Baker et al. (1974). All sample sites are from the Terevaka episode of volcanism. Site numbers correspond to site numbers in Table 1.

Terevaka. Initial interpretation was that the volcanic centers were separated by several million years time, as early K–Ar dating assigned ages of 2.5 million years for Poike lavas, 1.0 million years for Ran Kau and less than 0.24 million years for Terevaka activity (Clark and Dymond, 1977). Recent K–Ar dating on rocks from all three events indicate a much younger age for the entire island, with all samples yielding dates younger than 700,000 years (Miki et al., 1998). Lavas on Easter Island are transitional from island tholeiites to alkali basalt in composition, showing considerable fractionation, and suggestive of a shallow magma chamber under the island (Haase et al., 1997).

3. Previous paleomagnetic work

In 1970, 65 lava flows were collected from the Terevaka volcanic episode using a gasoline powered rock drill. Four samples, occasionally only three, were collected at each site, and oriented in situ with a Brunton compass. No sun compass readings were obtained. Sampling was restricted to the sea cliffs surrounding the island (Fig. 1). Due to lack of exposure in the interior of the island and the visual similarity between flows, few stratigraphic relationships are possible between the sampled flows. All samples were cut into two specimens, and standard paleomagnetic measurements were made on all samples (Isaacson and Heinrichs, 1976). Although, alternating field demagnetization techniques were employed on at least one sample from each site, many sites were found to be stable without any demagnetization. In the final data table published by Isaacson and Heinrichs (1976), 62% of the sites had no demagnetization, and only 10% of the sites received demagnetization of 20 mT or higher prior to site averaging. Sites that were demagnetized were subjected to a blanket demagnetization

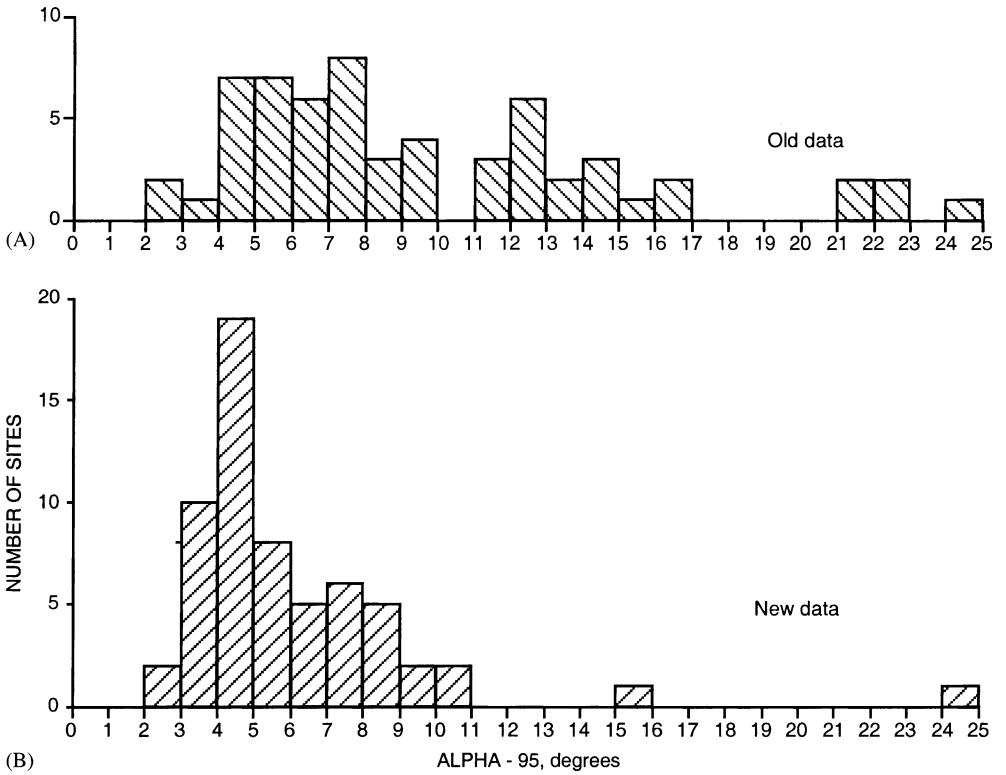


Fig. 2. Histograms of circle of confidence at the 95% level (α_{95}) values for each individual paleomagnetic site. Site means are determined by three or four samples, as indicated in Table 1. (A) Data from Isaacson and Heinrichs (1976); note that three sites lie beyond the right end of the graph with α_{95} values of 25.4, 32.2, and 48.2°; (B) data from the same sites after additional detailed alternating field magnetization.

level determined by the behavior of the single step-demagnetized specimen. Circles of confidence at the 95% level (α_{95}) around the site means ranged from 2 to 48° (Fig. 2A). There is a broad distribution of α_{95} values, with the majority of values being greater than 5°, and 25 sites with values greater than 10.0°. Mean directions were calculated for the entire population ($N = 63$) where $I = -43.4°$, $D = 357.9°$, $k = 30$, and $\alpha_{95} = 3.3°$. A selected subset eliminating any sites with VGP latitudes of less than 46° and any sites with less than four samples, leaving 51 sites. This population was used to calculate dispersion of the resulting VGPs of 12.8° with 95% error limits of 14.8 and 11.3° (Isaacson and Heinrichs, 1976). Results from this study indicated that the secular variation low observed around Hawaii by Doell and Cox (1972) was not as pronounced in the southeastern Pacific region.

Recently additional paleomagnetic work has been carried out on Easter Island (Miki et al., 1998). In this study, hand samples were collected from 45 sites representing all three volcanic episodes on the island. Stable magnetization was measured on 34 sites, all of which had normal polarity, agreeing with the 13 K–Ar determinations ranging from 0.7 to <0.12 Ma. A mean direction from this study is $I = -44.0°$, $D = 0.2°$, $k = 43$, and $\alpha_{95} = 3.8°$, agreeing well with the Isaacson and Heinrichs (1976) data. Angular dispersion for the Miki et al. (1998) data set is 11.8° with upper and lower error levels of 14.2 and 9.8°, respectively.

4. New paleomagnetic measurements

As the author had access to the original samples, data, and field notes, it was decided to treat the entire collection to detailed demagnetization procedures as now commonly done as part of any paleomagnetic study. All sites reported in Isaacson and Heinrichs (1976) were located for additional measurements except sites 50 and 54, and one additional unprocessed site (EC-57) was found for processing. Samples that had never been demagnetized were subjected to a minimum of seven steps of alternating field demagnetization at levels of 5, 10, 20, 40, 60, 80, and 100 mT. Principal component analysis was used to determine the resulting paleomagnetic directions. As can be seen in Fig. 3A and D, samples often showed straight line decay to the origin, or removal of a small overprint at 5 mT, and then simple decay. A few samples showed more resistant overprints (Fig. 3B), but these were usually removed by 20 mT. In the samples with slight overprints the result of removing them was to slightly steepen the inclination (Fig. 3D). For samples that had previously been blanket demagnetized, steps were begun at the blanket level and continued to 100 mT. As blanket levels were generally low (5 or 10 mT), a number of additional step demagnetizations were possible. In some cases companion specimens were used that had not been previously measured or demagnetized. A number of companion cores were also treated to thermal demagnetization, and in all cases thermal measurements produced similar characteristic directions to the alternating field results (Fig. 3C).

To accept cores for use in site mean determinations, several criteria needed to be met. These included the use of at least five steps in the demagnetization process to feed into the principal component analysis and the determination of the principal direction with a maximum angle of deviation (MAD) of less than 5°. Site mean directions were then determined from as few as three cores, but four wherever possible. Data were acceptable for secular variation considerations if α_{95} was <10°. These criteria are considerably less stringent than those employed in many modern studies focused on time averaged field directions (Johnson et al., 1998; Tauxe et al., 2000), but were necessary given the use of samples collected some 30 years ago.

5. Results

Using the new demagnetization data, characteristic remanence directions were determined for all samples. Site mean directions and poles calculated for the remeasured 62 sites are given in Table 1 and displayed in Fig. 4. As shown in Fig. 2B, the within-site scatter is considerably reduced with the additional demagnetization. The new measurements produce α_{95} values that range from 2 to 25°, with only six sites having values greater than 10.0°. In addition, half the population of 62 sites now has 95% confidence circles less than 5°. A mean direction for the entire data set of 62 sites is $I = -46.1°$, $D = 357.4°$ with $k = 23$ and $\alpha_{95} = 3.9°$. If sites are eliminated with circles of

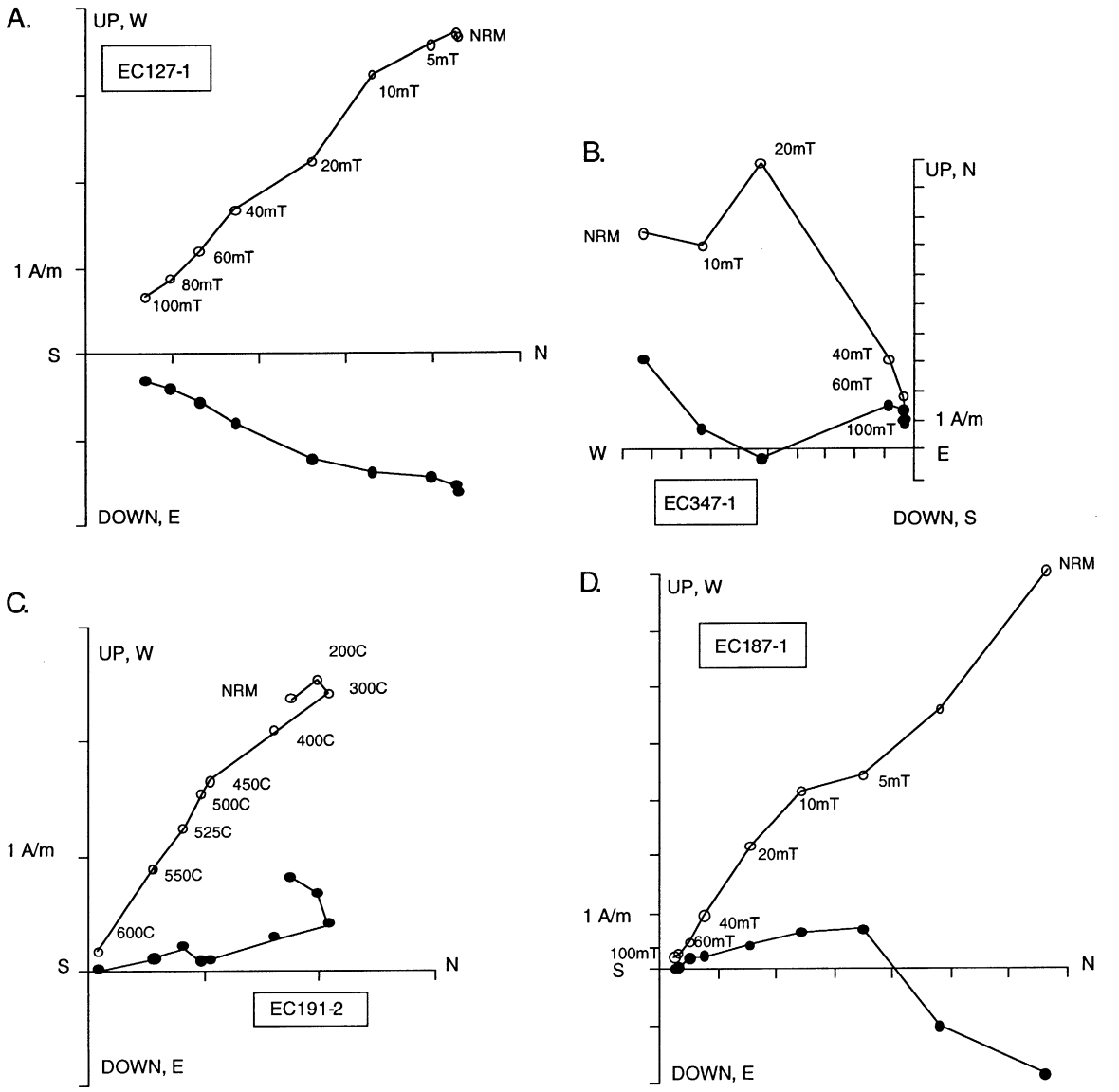


Fig. 3. Vector end-point diagrams showing typical demagnetization behavior for four samples from Easter Island. Open circles are projections onto the vertical plane; solid circles are projections onto the horizontal plane (A–D).

confidence greater than or equal to 10.0°, the mean directions are $I = -45.2°$, $D = 3573°$, $k = 24$ and $\alpha_{95} = 4.0°$ for $N = 55$ sites. Corresponding pole positions are 86.5°N, 125.4°E and 86.8°N, 139.8°E, respectively. Both pole positions barely include the rotation axis within circles of 95% confidence, and both show "far-sided" locations, as described by Wilson (1970).

Using the method described in McElhinny and Merrill (1975) and McElhinny and McFadden (1997) the paleosecular variation of this population was determined. Dispersion of the 55 VGPs with $\alpha_{95} < 10°$, taken with respect to the rotation axis, is 14.3°. When corrected for the within-site dispersion, this yields a value of 14.1° with upper and lower 95% error values of 16.2 and 12.4°, determined from the equations of

Table 1
Paleomagnetic directions, virtual geomagnetic poles and site statistics from lava flows, Terevaka volcanic complex, Easter Island

Flow	*N*	*I*	*D*	*k*	α_{95}	Plat	Plong
EC-1	3/3	−41.6	30.6	35	21.1	62.3	340.9
EC-2	4/4	−41.9	25.6	200	6.5	66.8	339.0
EC-3	3/4	−42.8	10.7	2667	2.4	80.1	329.4
EC-4	4/4	−50.0	354.0	348	4.9	83.5	123.8
EC-6	4/4	−35.1	346.4	116	8.6	75.3	189.4
EC-7	4/4	−48.2	4.0	748	3.4	85.9	12.4
EC-8	3/4	−41.8	9.8	389	6.3	80.7	324.0
EC-9	3/4	−54.3	333.6	696	4.7	66.1	135.1
EC-10	4/4	−54.9	337.1	509	4.1	68.8	131.8
EC-11	4/4	−49.9	2.7	252	5.8	85.7	37.8
EC-12	3/3	−53.7	355.6	234	8.1	81.9	97.5
EC-13	3/3	−42.5	2.6	694	4.7	86.6	294.1
EC-14	4/4	−51.3	316.3	272	5.6	53.7	142.6
EC-15	3/4	−44.1	13.1	151	10.1	78.2	337.5
EC-16	4/4	−39.2	13.1	107	8.9	77.2	321.4
EC-17	4/4	−57.3	350.0	75	10.6	76.3	105.7
EC-18	3/4	−50.3	354.4	1077	3.8	83.7	120.2
EC-19	4/4	−49.6	4.7	108	8.9	84.6	20.8
EC-20	3/4	−47.4	359.7	351	6.6	88.6	81.0
EC-21	4/4	−48.1	357.3	818	3.2	86.8	118.7
BC-22	4/4	−60.9	331.4	334	5.0	62.3	120.6
EC-23	3/4	−50.6	320.7	971	4.0	55.6	143.9
EC-24	4/4	−47.4	349.4	160	7.3	80.5	149.1
EC-25	4/4	−42.6	5.6	105	9.0	84.5	316.4
EC-26	3/4	−50.7	327.7	255	7.7	61.5	143.9
EC-27	3/4	−48.2	356.8	646	4.9	86.5	123.2
EC-28	4/4	−52.6	335.2	427	4.5	67.8	138.7
EC-29	4/4	−44.5	8.0	487	4.2	82.8	335.4
EC-30	4/4	−50.7	355.0	371	4.8	83.8	114.4
EC-31	4/4	−41.5	3.2	113	8.7	85.7	293.2
EC-32	4/4	−65.9	325.5	329	5.1	55.9	112.9
EC-33	3/4	−63.7	322.2	320	6.9	54.9	119.2
EC-34	4/4	−69.1	331.1	50	13.2	56.6	102.8
EC-36	3/4	−37.2	20.5	822	4.3	70.2	326.3
EC-37	4/4	−44.7	11.0	347	4.9	80.1	338.6
EC-38	4/4	−45.5	358.0	903	3.1	88.3	165.2
EC-39	4/4	−41.3	8.3	641	3.6	81.8	318.1
EC-40	4/4	−45.4	3.9	685	3.5	86.5	337.6
EC-41	4/4	−41.8	358.9	238	6.0	86.9	323.5
EC-45	4/4	−33.3	9.0	236	8.0	81.0	256.1
EC-46	3/4	−35.5	358.5	637	4.9	82.4	239.9
EC-47	4/4	−37.0	3.4	459	4.3	82.8	277.3
EC-48	4/4	−41.4	0.2	166	7.1	86.7	253.1
EC-49	4/4	−39.3	3.0	518	4.0	84.5	280.8
EC-51	3/4	−32.9	349.1	559	5.2	76.4	200.6
EC-52	3/4	−38.6	332.4	35	15.9	64.4	166.5
EC-53	3/4	−71.5	15.8	153	10.0	58.7	53.6
EC-55	4/4	−43.9	1.2	384	4.7	88.2	288.3
EC-56	4/4	−43.7	354.0	571	3.9	84.4	175.2
EC-57	3/4	−50.9	2.6	787	2.6	85.0	44.7
EC-58	4/4	−39.5	5.9	443	4.4	82.9	300.3
EC-59	3/4	−45.5	7.4	534	5.3	83.4	341.2
EC-60	4/4	−40.8	349.1	13	26.0	79.4	179.0
EC-62	4/4	−45.6	352.8	209	6.4	83.5	159.3
EC-63	4/4	−45.0	357.2	159	7.4	87.4	172.0
EC-64	4/4	−49.7	1.7	312	5.2	86.3	47.4
EC-65	4/4	−44.6	357.1	101	9.2	87.3	177.9
EC-66	4/4	−41.2	0.0	184	6.8	86.6	250.8
EC-67	4/4	−43.3	359.7	396	4.6	88.1	241.3
EC-68	4/4	−37.6	350.6	287	5.4	79.5	193.5
BC-69	4/4	−48.6	359.6	364	4.8	87.5	79.2
EC-70	4/4	−28.2	355.9	153	7.5	77.3	232.4
Mean	62/62	−46.1	357.4	23	3.9		
					3.4	86.5	125.4
Mean[a]	55/62	−45.2	357.3	24	4.0		
					3.3	86.8	139.83

N, number of samples used/number collected; *I*, inclination (in degrees); *D*, declination (in degrees); *k*, precision parameter; cone of confidence at 95% level; Plat and Plong, latitude and longitude of north magnetic pole.

[a] Mean omits seven sites with $\alpha_{95} > 10.0$.

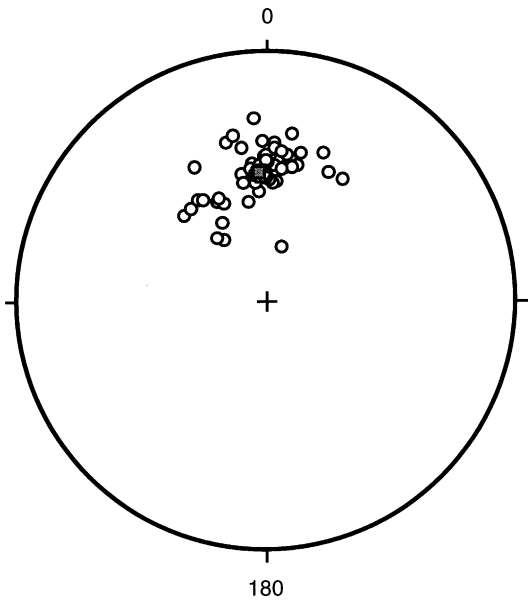


Fig. 4. Equal area net of all sites from Terevaka episode of volcanism, Baster Island. Grey square is mean direction for all sites.

Cox (1969b). The mean inclination is nearly identical to the predicted axial dipole inclination, so the inclination anomaly (ΔI) is only 0.5°.

6. Discussion

After detailed demagnetization studies and calculations of characteristic directions using principal component analysis, the mean directions from Easter Island lava flows are statistically very close to the directions published by Isaacson and Heinrichs (1976). This is true for either mean directions from all the flow measured, or for the populations screened for $\alpha_{95} < 10°$. In all cases the declinations lie within 1° of each other; inclinations for the newly demagnetized data are consistently steeper than the earlier directions by 2–3°. Angular dispersion of the poles for the two culled data sets differ by over a degree, but at the 95% level are concordant. The earlier data suffers from greater within-site dispersion which results in a lower field dispersion at the location.

The additional demagnetization done in this study does not greatly change the final mean direction nor the secular variation statistics. This is certainly due in part to the young age of the rocks measured, and the lack of overprints. Since all the sites are from flows in the later part of the Brunhes epoch, they have never been subjected to reversed fields or other directions vastly different from the present field. The oxide mineralogy and location environment is such that the samples have undergone little or no alteration that effects the magnetic directions. This is not always the case, as Carlut et al. (2000) report for lavas from the French West Indies which have distinctly different directions after extensive demagnetization than an earlier study without demagnetization.

When compared to another recent paleomagnetic study of Easter Island lavas (Miki et al., 1998), slight differences can be observed. Mean directions from the Miki study are slightly east of the results presented here, but are identical at the 95% level. Some variation is to be expected as their sampling included lavas from the older part of the island, whereas this study concentrates only on the youngest of the three volcanic centers. The angular dispersion calculated in their paper, $S_f = 11.8°$, is less than that presented here, but with considerable overlap when 95% error bars are considered. Some differences may be due to the sampling variations (drilled cores in this paper, hand samples from Miki et al. (1998) work) and the range of ages represented by the volcanic units studied. This paper presents data from the youngest volcanic sequence on Easter Island represented by over 60 distinct lava flows, while Miki et al. (1998) study sampled fewer flows ($N = 34$), but from the entire breadth of exposed volcanic activity.

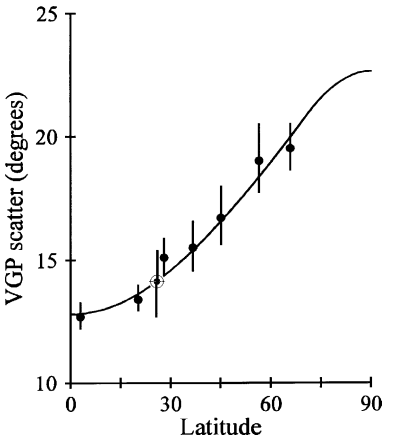


Fig. 5. Least-squares fit of Model G to both normal and reversed polarity scatter of virtual geomagnetic poles for the past 5 million years, from McFadden et al. (1991). Open circle is VGP dispersion for this set of Easter Island data; with 95% confidence limits after Cox (1969b).

The paleosecular variation determined here is plotted in Fig. 5 for comparison to the fit of paleosecular variation data from lava flows for the last 5 million years to Model G of McFadden et al. (1991). The data from Easter Island agrees with mean global values at similar latitude and with Model G. It does not indicate any region of low PSV in the Eastern Pacific as has been suggested for other parts of the Pacific Basin in the past by Doell and Cox (1972) and more recently by Shibuya et al. (1995).

Other anomalous features in the magnetic field have been suggested for the Pacific Basin region. Johnson and Constable (1998) observe large non-axial dipole contributions in the Pacific that have persisted over the past 10^2 to 10^6 years. Elmaleh et al. (2001) have used deep-sea cores and volcanic data to investigate the long-term behavior of the magnetic field in the southwest Pacific. A strong inclination anomaly of −10° is persistent in this area over the Brunhes epoch. Easter

Island, well to the east of the area in their study, does not show an inclination anomaly, supporting the contention of no anomalous field behavior in the Eastern Pacific.

7. Conclusions

After extensive demagnetization processing on a population of Brunhes lava samples with only scanty previous demagnetization, there is little change in paleomagnetic directions. The samples showed a slight tendency for inclinations to steepen during demagnetization, but declinations remained similar. The original mean direction (Isaacson and Heinrichs, 1976) and the complete demagnetized direction are similar at the 95% level. This lack of change of direction on additional demagnetization indicates that these lavas have collected few overprints during the short time since formation. A new calculation of paleosecular variation for Easter Island lavas yields a value similar to the global average for this latitude and thus does not support suggestions of a dipole low in this part of the Pacific basin.

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References

Baker, P.E., 1966. Preliminary account of recent geological investigations on Easter Island. Geol. Mag. 104, 116–122.

Baker, P.E., Buckley, F., Holland, J.G., 1974. Petrology and geochemistry of Easter Island. Contrib. Min. Petrol. 44, 85–100.

Bingham, D., Stone, D., 1972. Paleosecular variation of the geomagnetic field in the Aleutian Islands, Alaska. Geophys. J. R. Astron. Soc. 28, 317–335.

Carlut, J., Quidelleur, X., Courtillot, V., Boudon, G., 2000. Paleomagnetic and K–Ar dating of 0 to 1 Ma lava flows from La Guadeloupe Island (French West Indies): implications for time-averaged field models. J. Geophys. Res. 105, 835–849.

Clark, J.G., Dymond, J., 1977. Geochronology and petrochemistry of Easter and Sala Y Gomez islands: implications for the origin of the Sala Y Gomez Ridge. J. Volc. Geotherm. Res. 2, 29–48.

Cogne, J.P., Francheteau, J., Courtillot, V., Pito93 Scientific Team, 1995 large rotation of the Easter microplate as evidenced by oriented paleomagnetic samples from the sea floor. Earth Planet Sci. Lett. 136, 213–222.

Cox, A., 1969a. A Paleomagnetic study of secular variation in New Zealand. Earth Planet Sci. Lett. 6, 257–267.

Cox, A., 1969b. Confidence limits for the precision parameter *k*. Geophys. J. R. Astron. Soc. 17, 545–549.

Cox, A., 1971. Paleomagnetism of San Cristobal Island Galapagos. Earth Planet Sci. Lett. 11, 152–160.

Doell, R.R., Cox, A.V., 1971. Pacific geomagnetic secular variation. Science 71, 248–254.

Doell, R.R., Cox, A.V., 1972. The pacific geomagnetic secular variation anomaly and the question of lateral uniformity in the lower mantle. In: Robertson, E.C. (Ed.), The Nature of the Solid Earth. McGraw-Hill, New York, pp. 245–285.

Elmaleh, A., Valet, J.-P., Herrero-Bervera, E., 2001. A map of the Pacific geomagnetic anomaly during the Brunhes chron. Earth Planet Sci. Lett. 193, 315–332.

Haase, K.A., Stoffers, P., Garbe-Schonberg, C.D., 1997. The petrogenetic evolution of lavas from Easter Island and neighboring seamounts, near-ridge hot spot volcanoes in the SE Pacific. J. Petrol. 38, 785–813.

Isaacson, L.B., Heinrichs, D.F., 1976. Paleomagnetism and secular variation of Easter Island basalts. J. Geophys. Res. 81, 1476–1482.

Johnson, C.L., Constable, C.G., 1998. Persistently anomalous Pacific geomagnetic fields. Geophys. Res. Lett. 25, 1011–1014.

Johnson, C.L., Wijbrans, J.R., Constable, C.G., Gee, J., Staudigel, H., Tauxe, L., Forjaz, M., Salgueiro, 1998. $^{40}Ar/^{39}Ar$ ages and paleomagnetism of Sao Miguel lavas, Azores. Earth Planet Sci. Lett., 160, 637–649.

McElhinny, M.W., McFadden, P.L., 1997. Paleosecular variation over the past 5 million years based on a new generalized database. Geophys. J. Int. 131, 240–252.

McElhinny, M.W., Merrill, R.T., 1975. Geomagnetic secular variation over the past 5 million years. Rev. Geophys. Space Phys. 13, 687–708.

McElhinny, M.W., McFadden, P.L., Merrill, R.T., 1996. The myth of the Pacific dipole window. Earth Planet Sci. Lett. 143, 13–22.

McFadden, P.L., Merrill, R.T., McElhinny, M.W., Lee, S., 1991. Reversals of the earth's magnetic field and temporal variations of the dynamo families. J. Geophys Res. 96, 3923–3933.

McWilliams, M.O., Holocomb, R.T., Champion, D.E., 1982. Geomagnetic secular variation from ^{14}C dated lava flows on Hawaii and the question of the Pacific non-dipole low. Philos. Trans. R. Soc. Lond. A 306, 211–222.

Miki, M., Inokuchi, H., Yamaguchi, S., Matsuda, J., Nagao, K., Isezaki, N., Yaskawa, K., 1998. Geomagnetic paleosecular variation in Easter Island, the southeast Pacific. Phys. Earth Planet Int. 106, 93–101.

Naar, D.F., Hey, R.N., 1991. Tectonic evolution of the Easter microplate. J. Geophys. Res. 96, 7961–7993.

O'Connor, J.M., Stoffers, P., McWilliams, M.O., 1995. Time-space mapping of Easter chain volcanism. Earth Planet Sci. Lett. 136, 197–212.

Shibuya, H., Cassidy, J., Smith, I.E.M., Itaya, T., 1995. Paleomagnetism of young New Zealand basalts and longitudinal distribution of paleosecular variation. J. Geomag. Geoelectr. 47, 1011–1022.

Tauxe, L., Staudigel, H., Wijbrans, J.R., 2000. Paleomagnetism and $^{40}Ar/^{39}Ar$ ages from La Palma in the Canary Islands. Geochem. Geophys. Geosyst. 1, 2000GC000063.

Wilson, R.L., 1970. Permanent aspects of the earth's non-dipole magnetic field over upper tertiary times. Geophy. J. R. Astron. Soc. 19, 417–437.